

4.2 COORDINATE GEOMETRY

Coordinate Geometry is an *approach* to making investigations into, and solving problems from, the domain of geometric objects like straight lines, triangles, planes, cubes etc. The ***approach consists in representing geometric objects as algebraic expressions and equations*** like $ax + by + c = 0$ for a straight line, and $x^2 + y^2 = c$ for a circle etc. And then use algebraic tools for making inferences and solving problems. The process of translating geometric objects to algebraic expressions is achieved using ***coordinate systems***. There are several different coordinate systems, each more useful than the others for some specific purposes. We will discuss some of these briefly. However, before that we mention some synonymous and related terms, and a little bit of history.

Coordinate geometry is also called ***analytical geometry***, or ***cartesian geometry*** (in honour of *René Descartes*, 17th century French philosopher, and mathematician). Though evidence of the use of coordinates in geometry is found in the work of mathematicians from earlier periods, particularly of some Greeks and of the 11th century Parsian mathematician Omar Khayyam, yet the discipline in its current form was invented independently by two 17th century mathematicians viz. René Descartes and Pierre de Fermat.

Earlier form of geometry, which has been practiced for thousands of years, was retrospectively called since 17th century, ***synthetic geometry***, which uses, instead of algebraic tools, the *axiomatic method (based on proof techniques)* for proving results from a few basic properties called postulates/ axioms.

From our experience of school mathematics, we know that for 2-dimensonal space, ***cartesian coordinate system*** uses, as shown in Fig 4.5, two intersecting and mutually perpendicular ***real*** lines.

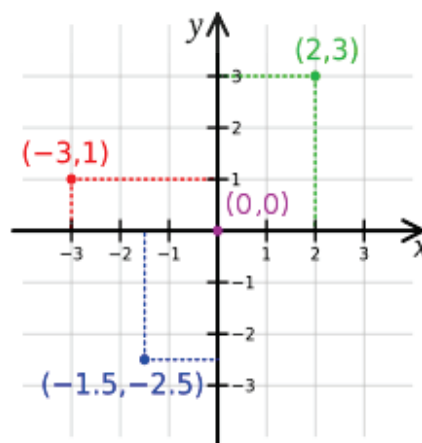


Fig 4.5: Two-Dimensional *Cartesian* Coordinate System

Though, we will be using mainly ***cartesian*** coordinate system, yet we should know the names of other coordinate systems which include ***polar coordinate system, cylindrical & spherical coordinate system, homogeneous coordinate system and Curvilinear coordinate system.***

4.2.1 Cartesian Coordinate System

In the rest of the discussion, unless mentioned otherwise, the term '**coordinate system**' will be used for **cartesian coordinate system**, which is also called **rectilinear coordinate system**.

In this coordinate system for 2-dimensional space, two mutually perpendicular intersecting lines (**called axes**), as shown in the figure below form basis of **rectangular co-ordinate system for 2-dimension**

- 1) The horizontal number line is called the **x-axis**.
- 2) The vertical number line is called the **y-axis**.

The **origin** is where the two lines/axes intersect. (See Fig. 4.6). It has four **quadrants** marked with Roman numerals.

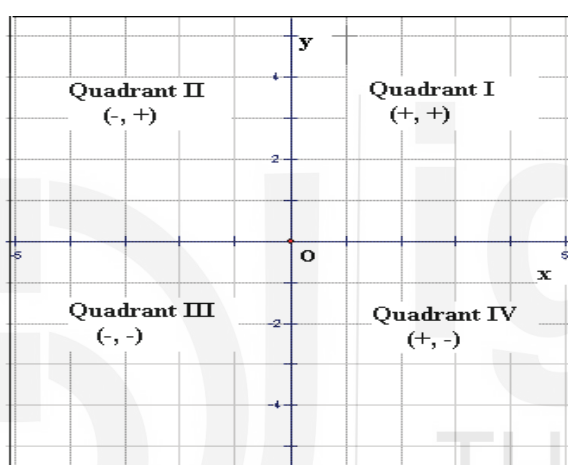


Fig. 4.6: Rectangular Co-ordinate System

Each point on the graph is associated with an **ordered pair**. At the **origin**, it is assumed that value of $x = 0 =$ value of y . When dealing with an (x, y) graph, the **x coordinate** is always written first and the **y coordinate** as second in the ordered pair **(x, y)**.

Normally, the values of the **independent** variable (generally the x -values) are placed on the horizontal axis, while the values of the **dependent** variable (generally the y -values) are placed on the vertical axis.

For a point $P(x, y)$, the x -value, called the **abscissa**, is the perpendicular distance of P from the y -axis, and the y -value, called the **ordinate**, is the perpendicular distance of P from the x -axis.

The values of x and y together, written as (x, y) are called the **co-ordinates** of the point P . The centre of the coordinate system (where the two lines intersect) is called the **origin**.

Such a formulation is called the "**rectangular**" **coordinate system** because the scale used along the x -axis is evenly spaced, as is the scale along the y -axis.

4.2.2 Translating Geometric Figures to Algebraic Equations

For translating a geometric figure, say, *a straight line*, we create a coordinate system around the object, or, equivalently, the geometric figure is embedded in a coordinate system as shown in Fig. 4.7. How and where to place the geometric figure in the coordinate system depends on the problem under consideration. Each point in Fig 4.7, has coordinates, as one of the points, say A, in the figure above, has coordinates (2, 0), with 2 as abscissa and 0 as ordinate, and another point, say B, has coordinates (0, -2).

In order to represent a geometric figure by an algebraic equation, we take an arbitrary point in Fig 4.7, assuming its coordinates as (x, y) with x as abscissa and y as ordinate. Then an *algebraic equation* representing the figure is some relationship (to be determined) between x and y.

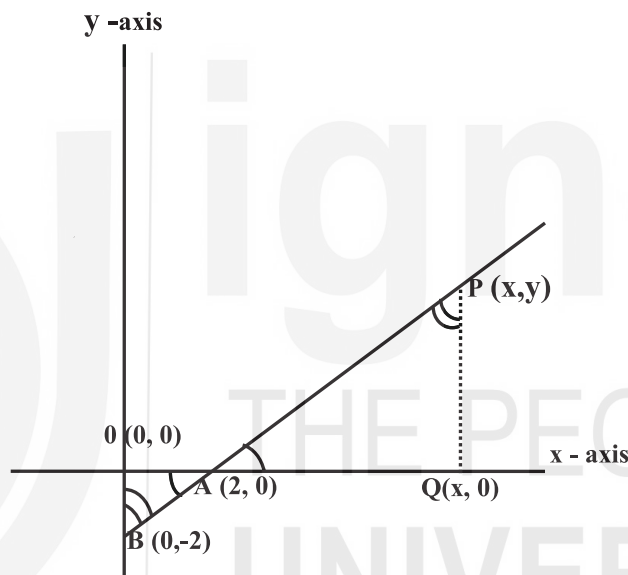


Fig. 4.7: Coordinate System

To *derive the algebraic equation* corresponding to Fig. 4.7, let us take P (x, y) outside the segment A (0, -2) and B (2, 0), and drop a perpendicular PQ on x-axis. Then triangles AOB and AQP are similar. From conventional/synthetic geometry, we know

$$|AO| / |AQ| = |BO| / |PQ|, \quad \dots (i)$$

Where, |X| denotes length of line segment X.

$$\text{Now } |AO| = 2, |AQ| = x - 2, |BO| = 2, |PQ| = y \quad \dots (ii)$$

From (i) and (ii) we get

$$2 / (x - 2) = 2 / y \text{ implying}$$

$$y = x - 2 \text{ is the algebraic translation of the Figure 4.7.}$$

Using similar method, we may obtain algebraic equations corresponding to the well-known curves, including the following standard algebraic forms

- $x^2 + y^2 = a^2$ for Circle

- $(x/a)^2 + (y/b)^2 = 1$ for Ellipse
- $y^2 = 4ax$, with $a > 0$ for Parabola
- $(x/a)^2 - (y/b)^2 = 1$ for Hyperbola
- $xy = c^2$ for Rectangular Hyperbola.

Also similarly, for solids of 3-dimensional space, corresponding algebraic forms may be obtained including the general equation of a **sphere** in the form

$(x - a)^2 + (y - b)^2 + (z - c)^2 = d$ with a , b , c , and d being constants, and $d > 0$

As the concern of this unit is how to draw a graph (a geometric figure) corresponding to a given algebraic equation, which is just the reverse of what we discussed just above. The discussion here was just for the sake of explaining how equations have been obtained for geometric figures.

Check Your Progress 1

- 1) Draw the graph of the equation $y^2 = x + 5$ and comment if it is a function.

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- 2) Find algebraic equation corresponding to a circle of radius r .

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4.3 GRAPHING LINEAR FUNCTIONS

A function $y = f(x)$, is called **linear** if the power of x in $f(x)$ is not more than 1. A linear *function* corresponds to a linear *equation* $Ax + By + C = 0$, in which degrees of x and y do not exceed 1, and A , B , C are constants.

To plot a graph of a function, we take, one by one, values in the domain (x_i) as independent variable x , and then find the corresponding values in the range y_i of the dependent variable y . It helps record the values of x and y while plotting the graphs. For example, consider a linear function depicting the relationship between price (x) and market demand (y) for a commodity. To plot the graph for evaluating the relationship, it is convenient to first prepare a table

called, ***t-chart***, comprising the values of x and y . If a function takes the form of $y = 7 - 5x$, then t -chart is as under:

Table 4.1: t -chart of $y = 7 - 5x$

x	$y = 7 - 5x$
-1	12
0	7
1	2
2	-3
3	-8

In general, we may have to take ***many*** values of the independent variable x , then find corresponding values of y . But in the *case of functions*, we need to take ***only two*** values of the independent variable x and calculate two corresponding values of the independent variable y . Thus, to draw the required graph, we ***need only*** the following part of the table

X	$y = 7 - 5x$
-1	12
0	7

The graph of such a function is as follows (refer Fig. 4.8):

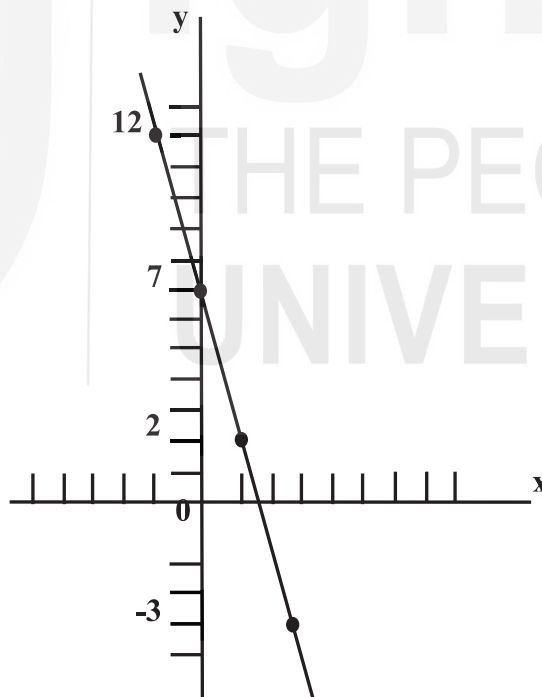


Fig. 4.8: Graph of Function $y = 7 - 5x$

The graph has ***x-intercept*** on a point where y is zero and ***y-intercept*** on a point where x is zero. For example, to find the x and y intercepts of the equation $3x + 4y = 12$ proceed as follows:

To find the x -intercept, set $y = 0$ and solve for x . That is, $3x + 4(0) = 12$ or, $x = 12/3 = 4$. Similarly, to find the y -intercept, set $x = 0$ and solve for y .

That is, $3(0) + 4y = 12$ or, $y = 12/4 = 3$. Thus, the x -intercept is $(4, 0)$ and the y -intercept is $(0, 3)$.

As a linear equation represents a straight line, and a straight line is specified by only two of its points. Hence, the graph of the line is as given in Fig 4.9.

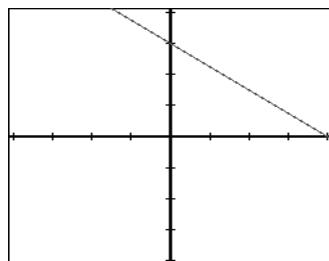


Fig. 4.9: Graph of $3x + 4y = 12$

4.3.1 Absolute Value Function

Absolute Value Function is given as $f(x) = |x|$ indicating that we need to consider modulus $|x|$ of a real number keeping it as the non-negative value without regard to its sign. That is, $|x| = x$ for a positive x , $|x| = -x$ for a negative x (in which case $-x$ is positive), and $|0| = 0$.

Thus, the function $y = |x|$, is defined by the following two equations

$$y = |x| = x \text{ for } x \geq 0, \quad \dots (1)$$

$$\text{and } y = |x| = -x \text{ for } x < 0 \quad \dots (2)$$

As each of (1) and (2) is a linear equation, each is represented by a straight line as shown in Fig. 4.10 below:

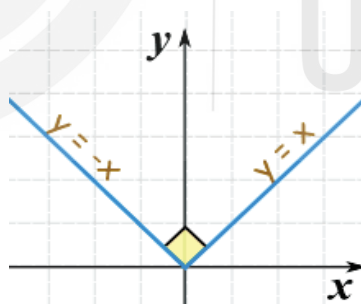


Fig. 4.10: Graph of Absolute Value

However, for some linear functions, the x -intercept & y -intercept method may not yield the required result, because the required curve may not intersect x -axis or y -axis.

Example: f is a function given by $f(x) = |x - 2|$. The y intercept is given by $(0, f(0)) = (0, |-2|) = (0, 2)$; x intercept is at the point $(2, 0)$ since we solve for $|x - 2| = 0$. But $0 = y = f(x) = |x - 2| = 0$ has no solution.

Since $|x - 2|$ is either positive or zero for $x = 2$; the domain of f is the set of all real numbers and the range of f is given by the interval $[0, +\infty)$. The corresponding graph is shown in Fig 4.11.

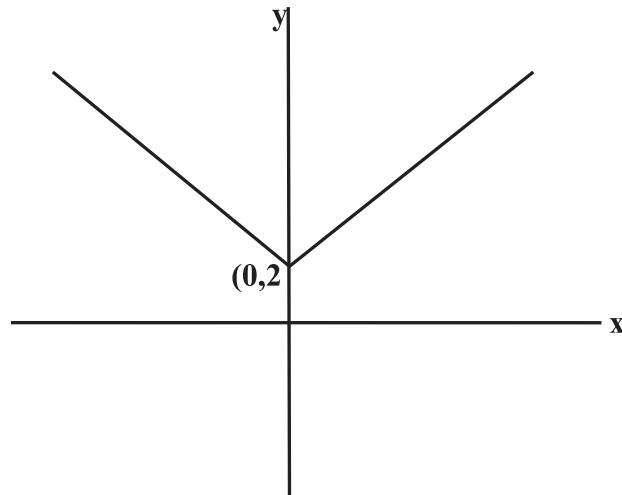


Fig. 4.11: Graph of $f(x) = |x - 2|$.

In general, for drawing a graph for a linear function, we need *just two* pairs (x_i, y_i) , $i = 1, 2$ of values of (x, y) .

4.3.2 Step Function

A step function (or staircase function) is a piecewise function containing all constant "pieces". The constant pieces are observed across the adjacent intervals of the function, as they change value from one interval to the next. A step function is discontinuous (not continuous) as you can see in the Fig. 4.12. You cannot draw a step function without removing your pencil from your paper.

Example of a step function: For definition of our particular step function $y = f(x)$,

First, we write every $x = \text{Int-}x + \text{Fraction-}x$, where $\text{Fraction-}x \geq 0$.

E.g., for $x = 5.46$, $\text{Int-}x = 5$, and $\text{Fraction-}x = .46$

For $x = -3.87$, $\text{Int-}x = -4$, and $\text{Fraction-}x = .13$

Then we define $y = f(x) = \text{Int-}x$

Thus, $f(5.46) = 5$, and $f(-3.87) = -4$.

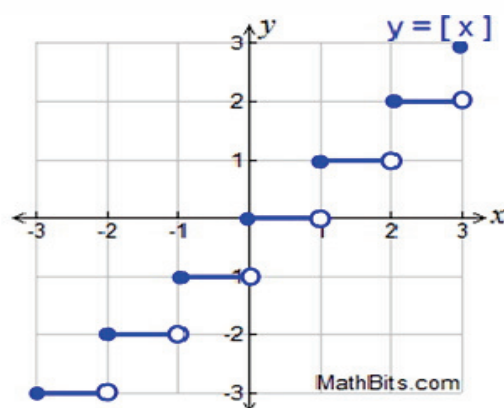


Fig. 4.12: Graph of Step Function

Note: The *domain* of a function is the set of all inputs for which the function is defined. For piece wise functions, the domain is union of all the individual cases. The *range* of a function is the set of all the possible function outputs. For piece wise functions, this is the union of the range so fall the individual cases.

Check Your Progress 2

- 1) Draw the graph of $y = (-5/3)x - 2$.

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- 2) Find when $f(x) = 2|3x+2| - 12$ crosses the x-axis.

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- 3) A truck rental company, charges a rental fee of Rs. 25 plus Rs. 0.30 per mile for driving between 0 and 500 miles for the standard -sized truck. For the same truck, the company charges Rs.100 plus Rs. 0.15 per mile for driving more than 500 miles, but less than 1000 miles. Write a step function that represents the situation.

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4.4 GRAPHING NON-LINEAR FUNCTIONS

Determining the nature of the functions that produces non-linear type curves requires additional considerations compared to those of linear types. When you graphed straight lines, you only needed two points to graph the line, though you generally plotted three or more points just to be on the safe side. However, three points will almost certainly *not* been ought points for graphing curve so f quadratic or cubic functions. We discuss the caution to be exercised for sketching the graphs of such function.

4.4.1 Even and Odd Functions

A function f is **even** if its graph is symmetric with respect to the $y - ax$ is. This criterion can be stated algebraically as follows: f is even if $f(-x) = f(x)$ for all x in the domain off. For example, if you evaluate at 3 and at -3, then you will get the same value, if f is even. Geometrically, such functions are symmetric about the origin.

We have seen that for an even function $f(x)=f(-x)$ for all x , there is symmetry about y-axis. The graph, in this case is shown in Fig 4.13.

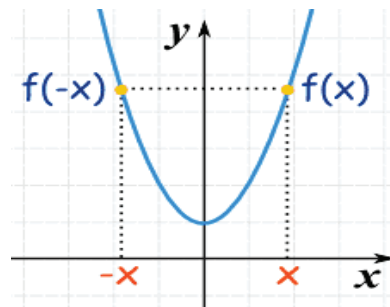


Fig. 4.13:Graph of Even Function

A function is "odd" when $f(x) = -f(-x)$ for all x and symmetry is observed from the point of origin (Fig. 4.14).

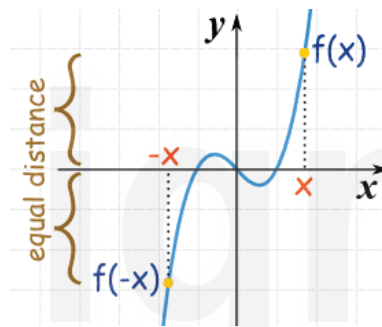


Fig. 4.14:Graph of Odd Function

Note: The graph based on $f(x)=x^3-x$ is an odd function

4.4.2 Quadratic Functions

Quadratic functions help describing demand, cost, revenue and profit which you will find while covering microeconomic analysis. The general technique for graphing quadratics is the same as **for graphing linear functions**. However, since quadratics graph as curvy lines (for example, parabolas, ellipses etc.), rather than the straight lines generated by line equations, there are some additional considerations.

The most basic quadratic is $y = x^2$. When you graphed straight lines, you only needed two points to graph your line, though you generally plotted three or more points just to be on the safe side. However, three points will almost certainly *not* be enough points for graphing a quadratic. For example, compute these three points:

Table 4.2: T-chart of $y=x^2$

x	$y=x^2$
0	0
1	1
2	4

Then, based only on this experiment, the plotted graph will produce a straight line. Thus, the graph is not the correct presentation of the function. More points need to be considered. By extending the above table with more points we construct Table 4.3.

Table 4.3: Extension with more points

X	-3	-2	-1	0	1	2	3
Y	9	4	1	0	1	4	9

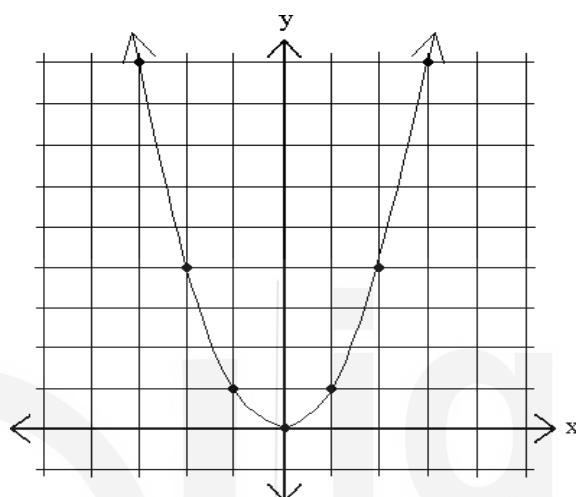


Fig. 4.15: Graph of Equation $y=x^2$

We observe that the graph in Fig 4.15 is a parabola. Note that the parabola does not have a constant slope. In fact, as x increases by 1, starting with $x = 0$, y increases by 1, 3, 5, 7, . . . As x decreases by 1, starting with $x=0$, y again increases by 1,3,5,7,

Consider a quadratic function of the form: $f(x) = ax^2 + bx + c$ with $a \neq 0$. The graph of such a function takes one of the two general forms shown in Fig. 4.16, depending on the sign of a .

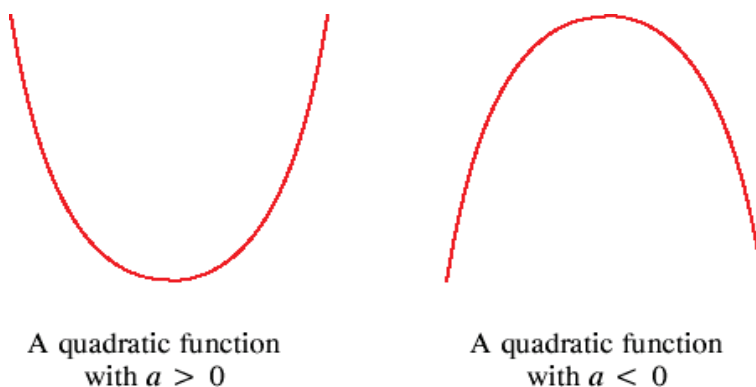


Fig. 4.16: Graph of Quadratic Function